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April 30, 1986
SUPERSEDING
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April 26, 1977

FEDERAL SPECIFICATION

INSECT REPELLENT, CLOTHING AND PERSONAL APPLICATION

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for two types of diethyltoluamide insect repellent.

1.2 Classification.

1.2.1 Types, solution concentrations, and sizes. The repellents shall be one grade and of the following types, solution concentrations, and sizes, as specified (see 6.2):

Type I	- Diethyltoluamide, technical 95 percent
Size 1	- 2 fluid ounces
Size 2	- 1 gallon
Type II	- Diethyltoluamide solutions
Concentration A	- 75 percent diethyltoluamide in ethanol
Concentration B	- 50 percent diethyltoluamide in ethanol
Concentration C	- 30 percent diethyltoluamide in ethanol

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6840

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Size 1	- 1/2 fluid ounce (only applicable to concentrations A and C)
Size 2	- 2 fluid ounces
Size 3	- 1 gallon

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

O-E-760	- Ethyl Alcohol (Ethanol); Denatured Alcohol, Proprietary Solvent and Special Industrial Solvents
TT-E-485	- Enamel, Semigloss, Rust-Inhibiting
PPP-B-601	- Boxes, Wood, Cleated Plywood
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-C-96	- Cans, Metal, 28 Gage and Lighter

Federal Standards:

FED-STD-123	- Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

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Military Specifications:

MIL-B-26701	- Bottles, Screw Cap and Carboys, Polyethylene, Plastic

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage

(Copies of military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Laws and Regulations:

Bureau of Alcohol, Tobacco and Firearms, Department of the Treasury

Formulas for Denatured Alcohol (27 CFR Part 21)

(Application for copies should be addressed to the Bureau of Alcohol, Tobacco, and Firearms, Department of the Treasury, 1200 Pennsylvania Avenue, Washington, DC 20226.)

Environmental Protection Agency

Regulations for the Enforcement of the Federal Insecticide, Fungicide and Rodenticide Act (40 CFR Part 162)

(Application for copies should be addressed to the Environmental Protection Agency, 401 M Street SW, Washington, DC 20460.)

(The Code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials (ASTM)

D 1209 - Color of Clear Liquids (Platinum-Cobalt Scale)
D 1364 - Test Method for Water in Volatile Solvents (Fischer Reagent Titration Method)
D 1639 - Test Method for Acid Value of Organic Coating Materials
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Glass Container Manufacturers Institute (GCMI)

GCMI Recommended Glass Finish 410

(Application for copies should be addressed to the Glass Container Manufacturers Institute, 1800 K Street N.W., Washington, DC 20006.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Diethyltoluamide. The diethyltoluamide used in formulating insect repellent, clothing and personal application, shall be technical grade and shall contain N, N-diethyl-m-toluamide of not less than 95 percent purity by weight and shall meet the requirements in table I. The remainder shall consist almost entirely of ortho or para isomers of N, N-diethyltoluamide or a mixture of both isomers as indicated by the infrared absorption spectra.

TABLE I. Chemical and physical properties

Properties	Requirement
N, N-diethyl-m-toluamide	95 percent, minimum
Refractive index at 25°C	1.5200-1.5240
Specific gravity (25/25°C)	0.992-0.999

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TABLE I. Chemical and physical properties - Continued

Properties	Requirement
Acid number	0.15 maximum
Moisture	0.5 percent maximum
Color	Not darker than Platinum Cobalt Color Standard number 100
Odor	Near odorless, except for characteristic odor of diethyltoluamide for use in type II repellent

3.1.2 Denatured alcohol in type II repellent. Denatured alcohol shall be grade III, U.S.P. 94.9 percent by volume, of O-E-760, denatured in accordance with Code of Federal Regulations 27 CFR 21, Formula 40.

3.2 Composition. These insect repellents must be registered with the U.S. Environmental Protection Agency (EPA) in accordance with the Federal Insecticide, Fungicide and Rodenticide Act (FIFRA).

3.2.1 Type I. Type I shall consist of diethyltoluamide conforming to the requirements of 3.1.1.

3.2.2 Type II.

3.2.2.1 Concentration A. The material shall consist of 75 percent, by weight minimum, of diethyltoluamide specified in 3.1.1 and the remainder denatured alcohol specified in 3.1.2.

3.2.2.2 Concentration B. The material shall consist of 50 percent, by weight minimum, of diethyltoluamide specified in 3.1.1 and the remainder denatured alcohol specified in 3.1.2.

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3.2.2.3 Concentration C. The material shall consist of 30 percent, by weight minimum, of diethyltoluamide specified in 3.1.1 and the remainder denatured alcohol specified in 3.1.2.

3.3 Label. The label information shall be legibly and neatly printed, stenciled, lithographed, or silk-screened directly on the container if compatible with the type of material of which the container is made. As an alternative, the label information shall be legibly and neatly printed on Olive Drab paper stock conforming to Color No. 24087 of FED-STD-595. The label shall be sealed to the container with water-resistant adhesive. In all cases, the ink used shall be black, regardless of how the label information is applied to the bottle. The label shall not smear or be loosened from the container when tested as specified in 4.3.7.

3.4 Workmanship. The products shall be clean, uniform, and free from any defects that might impair their utility.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. In addition to testing provisions contained in subsidiary specifications, drawings, or standards, testing shall be performed on components listed in table II for characteristics noted. The lot size shall be expressed in pounds of diethyltoluamide. The sample unit for testing shall be a one-pound composite obtained by randomly selecting and thoroughly mixing equal portions from all containers in the sample. The composite sample shall be placed in a clean, airtight, watertight glass container with suitable caps, covered and sealed to prevent any moisture change. All test reports shall contain the individual values used in expressing the final result. The lot shall be unacceptable if the composite fails to meet any test requirement specified. The sample size shall be in accordance with the following:

Lot size (pounds)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. Instructions for testing of materials and components

Characteristic	Specification reference: Rqmt.	Specification reference: Test method	Requirements applicable to composite sample	Number of determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest
N, N-diethyl-m-toluamide	Table I	4.3.1	X	2	-	1.0 percent
Refractive index at 25°C	Table I	4.3.2	X	2	-	0.0001

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TABLE II. Instructions for testing of materials and components - Continued

Characteristic	Specification reference: Rqmt.	Specification reference: Test method	Requirements applicable to composite sample	Number of determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Specific gravity (25/25°C)	Table I	4.3.3	X	2	-	0.001
Acid number	Table I	4.3.4	X	2	-	mg
Moisture	Table I	4.3.5	X	2	-	1.0 percent
Color (Platinum-Cobalt)	Table I	ASTM D 1209	X	1	X	-
Odor	Table I	4.3.6	X	1	X	-

4.2.2 Inspection of the end item. The end item shall be examined for the defects in 4.2.2.1, 4.2.2.2, and 4.2.2.3. The lot size shall be expressed in units of filled unit containers of the same type and size.

4.2.2.1 Examination of the unit container. The unit container shall be examined for the defects listed in table III. The sample unit for this examination shall be one filled unit container. The inspection level shall be I, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

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TABLE III. Examination of the unit container

Examine	Defect
Construction	Not as specified (see 5.1)
Closure	Not as specified (see 5.1)
Markings (label)	Missing, incomplete, incorrect, illegible, not as specified (see 3.3)
Workmanship	Dirty; any leaking of contents; any other defect affecting serviceability

4.2.2.2 Examination of contents of the unit container. The contents of the unit container shall be examined for workmanship. The sample unit for this examination shall be the contents of one filled unit container. The sample shall be transferred into a clear glass container and after standing for three hours shall be examined for foreign particles visible to the naked eye. The presence of visible foreign particles shall be considered a defect. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 1.0.

4.2.2.3 Examination for average net contents. The end item shall be examined for defects in net contents. The sample unit for this examination shall be one filled unit container. The inspection level shall be S-2. The lot shall be unacceptable if the average net content per container for all sample units examined is less than the quantity specified.

4.2.3 Testing of the end item. The end item shall be tested for the characteristics specified in table IV. The lot size shall be expressed in cans or bottles, of repellent of the same type and size, as applicable. The sample unit shall be one filled and sealed can or bottle. In addition, the contractor shall submit a certificate of compliance with each shipment or lot citing conformance to the requirements in 3.2.2.1, 3.2.2.2, and 3.2.2.3 when type II insect repellent is being procured. All test reports shall contain the individual values used in expressing the final result. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be in accordance with the following:

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Lot size (containers)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV. Testing of the end item

Characteristic	Requirement	Test method
Label resistance to water	3.3	4.3.7.1
Label resistance to insect repellent	3.3	4.3.7.2

4.2.4 Packaging inspection. An examination shall be made for the defects listed in table V to determine that preservation, packing, and marking comply with section 5 requirements. The lot size shall be the number of shipping containers in the inspection lot. The sample unit shall be one shipping container fully packed. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

TABLE V. Packaging inspection

Examine	Defect
Markings (unit packs, intermediate packs, and shipping containers)	Incorrect; incomplete; illegible; omitted; of improper size, sequence or method of application; not in accordance with contract requirements
Material	Any nonconforming component; component missing, damaged, or otherwise defective
Workmanship	Inadequate application of components Bulged or distortion of intermediate boxes or shipping containers

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TABLE V. Packaging inspection - Continued

Examine	Defect
Contents (intermediate packs)	Number of unit packs is more or less than specified
Weight (shipping container)	Weight of contents exceeds 65 pounds

4.3 Test Methods.

4.3.1 Determination of N, N-diethyl-m-toluamide.

4.3.1.1 Apparatus.

a. Double-beam infrared spectrophotometer (Perkin-Elmer Model 21 or equivalent).

b. Two equivalent absorption cells, infrared, approximately 0.4 mm, with rock salt windows.

4.3.1.2 Solvent.

Carbon disulfide - ACS reagent grade.

4.3.1.3 Reference standard. (see 6.3)

N, N-diethyl-m-toluamide (b.p. 111°C at 1 mm)
(ref. index n_D^{25} 1.5206)

4.3.1.4 Preparation of standard curve.

a. Prepare, in volumetric flasks, carbon disulfide solutions of N, N-diethyl-m-toluamide of approximately 20, 40, 60, and 80 mg/mL. At the peak absorbance of 14.18 microns, absorbance units will fall roughly between 0.2 and 0.8.

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b. Fill the reference cell with CS_2 and the sample cell with each of the standard solutions in turn, and record the spectrum. The spectrum may be scanned rapidly except for the region 12 to 15 microns, where a normal speed should be used. The absorbance difference (ΔA) is the difference between the absorbancies of the CS_2 solution and the blank at the peak at 14.18 and the minimum at 14.48, that is:

$$\Delta A = (A_{14.18} - A_{14.48})_{\text{(meta-isomer)}} - (A_{14.18} - A_{14.48})_{\text{(CS_2 blank)}}$$

c. Prepare a standard curve by plotting the absorbance difference (ΔA) versus concentration (mg/mL) of meta-isomer for each of the standard solutions. Determine the N, N-diethyl-m-toluamide content of the formulation from this curve.

4.3.1.5 Procedure.

a. Weigh accurately a sufficient amount of sample to give a 40 to 70 mg/mL concentration (with the 0.4 mm cell) of the meta-isomer in CS_2. Record the infra-red absorption using the same absorption cell conditions described under calibration.

b. Calculate ΔA as described and determine the milligrams per milliliter of meta-isomer from the standard curve. A standard sample should be run each day to check the calibration of the instrument.

c. Final calculations:

$$\frac{\text{mg/mL of isomer from calibration curve X 100}}{\text{mg/mL of sample in solution}} = \text{percent meta-isomer}$$

4.3.2 Refractive index. The refractive index shall be determined by any method accurate to the fourth decimal point.

4.3.3 Specific gravity. The specific gravity shall be determined by any method accurate to the third decimal point.

4.3.4 Acid number. The acid number shall be determined in accordance with ASTM D 1639.

4.3.5 Moisture. Moisture shall be determined by direct determination of water by titration with a solution of iodine, sulfur dioxide, and pyridine in methanol (Karl Fischer Reagent). The procedure shall be as specified in ASTM D 1364.

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4.3.6 Odor. Note presence of odor other than diethyltoluamide after saturating cloth or paper.

4.3.7 Labels.

4.3.7.1 Resistance to water. The container shall be completely immersed in water at 70° to 80°F for one hour. Remove the container and rub fingers over the face of the label to determine if the label either smears or loosens to the point that it can be moved. Any smearing or loosening of the label constitutes a failure.

4.3.7.2 Resistance to insect repellent. The container shall be completely immersed in the type and concentration of repellent contained therein at 70° to 80°F for one hour. Remove the container and rub fingers over the face of the label to determine if the label either smears or loosens to the point that it can be moved. Any smearing or loosening of the label constitutes a failure.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packing. One-half fluid ounce, two fluid ounces, and one-gallon quantities of types I and II repellent shall be unit packed in polyethylene bottles, glass bottles, or metal can containers in specified sizes (see 6.2).

5.1.1.1.1 One-half ounce polyethylene bottle. One-half fluid ounce of type II (concentration A or C, as applicable) repellent shall be unit packed in a polyethylene bottle as specified in 5.1.1.1.2, except the bottle shall have approximate outside dimensions of 2 5/8 inches overall height, with a major axis of 1 5/16 inches and minor axis of 11/16 inch.

5.1.1.1.2 Two-ounce polyethylene bottle. Two fluid ounces of repellent shall be unit packed in a polyethylene bottle conforming to MIL-B-26701, except for the following deviations: Each polyethylene bottle shall conform to color No. 24087 Olive Drab of FED-STD-595. The Olive Drab color shall be produced by the use of inorganic pigments. The bottle shall be oval around the vertical axis and shall be formed in one piece with an 18 - 410 medium (depth), continuous thread concealed bead finish in accordance with GCMI Recommended Glass Finish 410. The approximate dimensions of the bottle shall be 3 3/4 inches over-all height, with a major axis of 1 15/16 inches and a

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minor axis of 1 inch. The minimum wall thickness shall be 0.01 inch. The empty bottle weight, less cap and plug, shall be 9 grams ± 1/2 gram. The inside dimension of the finish shall be 0.476 ± 0.006 inch. Each finish shall be provided with a 0.500 ± 0.005 inch diameter tube well spray-orifice plug. Each plug shall have a 0.030 ± 0.005 inch diameter spray-orifice. Each bottle shall be provided with a cap made of black phenolic or melamine thermosetting plastic. The cap shall match the 18 - 410 finish of the bottle. The cap shall be furnished with a securely fitted liner 0.040 inch in thickness and faced with 0.0015 inch thick tin or aluminum foil. When the cap is completely screwed onto the bottle, the plug shall fit firmly against the liner to ensure sealing against leaking. There shall be not less than 1/8 inch clearance between the bottom of the skirt of the cap and any portion of the shoulder area of the bottle.

5.1.1.1.3 Two-ounce glass bottle. Two fluid ounces of repellent shall be unit packed in a glass bottle that shall be rectangular in shape, or the applicator type, and shall be made of heavy flint glass with a threaded glass neck. The bottles shall have a constricted orifice 13/64 inch in diameter. A tolerance of plus or minus 1/32 inch will be permitted. Each bottle shall be provided with a cap made of black phenolic or melamine thermosetting plastic. The cap shall match the finish of the bottle and shall be sufficiently shallow to allow a tension closure (i.e., when the cap is tightly closed, its edge shall not contact any rim or shoulder of the bottle). The cap shall be furnished with a securely fitted paperboard liner 0.040 inch in thickness and faced with a 0.0015 inch thick tin or aluminum foil.

5.1.1.1.4 Two-ounce metal can. Two fluid ounces of repellent shall be unit packed in a metal can that shall be liquid tight, oblong-base, screw cap style, approximate dimensions of 2 1/4 by 1 1/8 inch body height over double seams 2 9/16 inches overall height, with soldered side seam and compound lined double seamed top and bottom. Can top shall be fitted with a 5/8-inch threaded nozzle, perforated with a pouring opening approximately 13/64 inch in diameter and closed with a curled edge, knurled, continuous thread screw cap fitted with a suitable liner pad and facing. Can body, bottom, and screw cap shall be coated outside with an Olive Drab corrosion-resistant enamel in accordance with TT-E-485. Screw caps shall be coated inside with a corrosion-resistant enamel which will not be affected by the product. Cans shall be made throughout with 1.25-pound base box of 112 sheets, 14 by 20 inches hot dipped tinplate.

5.1.1.1.5 One-gallon metal can. One gallon of repellent shall be unit packed in an oblong metal can conforming to type 5, class 4 of PPP-C-96. The exterior coating of each can shall conform to plan B. The screw neck of each can shall be fitted with a pryout tamper proof inner seal. Each can shall be

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provided with a formed bridge type handle, constructed of the same material as the can, securely affixed to the top of the can. Each can shall be closed as specified in the appendix of PPP-C-96.

5.1.1.2 Intermediate packing. Insect repellent of one type and concentration only, unit packed as specified in 5.1.1.1, shall be placed in a snug-fitting fiberboard box in the unit pack size, arrangement, and quantity as specified in table VI. The fiberboard box shall conform to style RSC, type CF, (variety SW), class domestic, grade 200 of PPP-B-636. Each box containing glass bottles and metal cans, except the one-gallon, shall be provided with snug-fitting, full-height, half-slotted style interlocking partitions. Each box containing unit pack sizes of one-half fluid ounce or two fluid ounces shall be provided with a tier separator pad. In addition, each box containing glass bottles shall be provided with top and bottom pads. The partitions and pads shall be constructed of the same material as the box. The bridge type handles on the one-gallon cans shall be protected from damage by appropriate die-cuts and scored sheet of the same material as used in the fiberboard box. Each box shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

TABLE VI. Size, arrangement, and quantity of unit packs within intermediate box

Unit pack size	Number of rows	Number in rows	Number of tiers	Total (each)
One-half fluid ounce	3	8	2	48
Two fluid ounces	3	8	2	48
One-gallon	2	2	1	4

5.1.2 Commercial. Insect repellent shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

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5.2.1 Level A packing. Insect repellent of one type, concentration and pack size only, preserved as specified in 5.1, shall be packed in a snug-fitting cleated-plywood shipping container conforming to overseas type, type 2 load of PPP-B-601. The weight of contents of each shipping container shall not exceed 65 pounds.

5.2.2 Level B packing. Insect repellent of one type, concentration, and pack size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636. The weight of contents of each fiberboard container shall not exceed 65 pounds.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with method III as specified in the appendix of PPP-B-636.

5.2.2.2 Special packing. When specified by Civil agencies (see 6.2), intermediate packing specified in 5.1.1.2 does not require additional packing.

5.2.3 Commercial packing. Insect repellent of one type, concentration, and pack size only, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or purchase order, unit and intermediate packs and shipping containers shall be marked in accordance with FED-STD-123 or ASTM D 3951, as applicable.

5.3.2 Military requirements. In addition to any special marking required by the contract or purchase order, unit and intermediate packs and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 Special marking. Each intermediate pack shall be marked with the batch number. Each intermediate pack and shipping container containing glass bottles shall be marked to indicate the containment of fragile items.

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5.3.4 Labeling information for each size container. Labeling shall be in compliance with the requirements of the Federal Insecticide, Fungicide and Rodenticide Act as amended. Labeling shall contain, as a minimum, the directions for use and precautionary information as shown in 5.3.4.1 and 5.3.4.2. Labels shall be approved by the EPA, Washington, DC 20460.

5.3.4.1 One-half and two-ounce containers. The directions and caution must be in accordance with and approved by the EPA. The following is an example.

EPA REGISTRATION NUMBER

EPA ESTABLISHMENT NUMBER

INSECT REPELLENT - (APPLICABLE TYPE)

(Federal Specification O-I-503, Insect Repellent, Clothing, and Personal Application)

Contents one-half fluid ounce or two fluid ounces, as applicable

ACTIVE INGREDIENTS

N, N-diethyl-meta-toluamide	%
Other diethyltoluamides	%

INERT INGREDIENTS

Denatured ethanol	%
	100%

DIRECTIONS

This product will repel mosquitoes, biting flies (to include black, deer, horn, horse, snipe, greenhead, and stable flies) biting midges, ticks, chiggers, gnats, and fleas, when applied to exposed skin.

Skin Application: Shake about 12 drops into one hand, rub hands together, and apply thoroughly in a thin layer to all areas of exposed skin, much as in washing. Sufficient repellent should be applied to give a uniform film. APPLY WITH CAUTION AROUND EYES AND MOUTH.

Clothing Application: When clothing cannot be impregnated with bulk repellent, apply by pouring about 12 drops into one hand, rub the hands together, and rub lightly on the socks and other clothing. Make liberal

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applications along all openings of the clothing, such as inside the neck band and the fly and bottom of trousers, and where clothing fits tightly, as across the shoulders, elbows, hips, and knees. Repeat smearing process several times until about 2 ounces of repellent have been used. One treatment of clothing is effective for several days if repellent is not washed away. Clothing may be treated the day before wearing. If irritation develops, discontinue wearing treated clothing.

CAUTION

KEEP OUT OF REACH OF CHILDREN - FOR EXTERNAL USE ONLY - HARMFUL IF SWALLOWED - AVOID CONTACT WITH EYES AND MOUTH

DO NOT SPRAY DIRECTLY ON FACE

PRODUCT MAY AFFECT LACQUERS, PAINTS, PLASTICS AND SYNTHETIC FABRICS

Manufacturer's name and address

5.3.4.2 One gallon containers. The directions and caution must be in accordance with and approved by the EPA. The following is an example.

EPA REGISTRATION NUMBER

EPA ESTABLISHMENT NUMBER

INSECT REPELLENT - (APPLICABLE TYPE)

(Federal Specification O-I-503, Insect Repellent, Clothing, and Personal Application)

Contents one gallon

ACTIVE INGREDIENTS	
N, N-diethyl-meta-toluamide	%
Other diethyltoluamides	%
INERT INGREDIENT	%
	100%

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DIRECTIONS

Apply to clothing by use of insecticide sprayer (hand or electric paint sprayer or knapsack type sprayer) using a coarse spray. Usually 2 to 2-1/2 fluid ounces per suit is required to give protection. Avoid prolonged breathing of mist.

For protection against mosquitoes, biting flies (to include black, deer, horn, horse, snipe, greenhead, and stable flies), biting midges, ticks, chiggers, gnats, and fleas. Pay special attention to those parts of the suit where biting through clothes occurs - shoulders, elbows, seat of trousers, front of thighs, knees, and socks.

For protection against chiggers (mites), fleas and ticks, pay special attention to the socks, shoe tops and fly, trouser leg cuffs, trouser fly, and waistband and the front fly and cuff fly of the shirt.

FOR EXTERNAL USE ONLY

CAUTION: Keep out of reach of children; harmful, if swallowed; avoid contact with eyes and mouth. The spraying of suits does not protect the exposed skin.

DO NOT SPRAY DIRECTLY ON FACE

THIS MATERIAL AFFECTS LACQUER, PAINT, CERTAIN PLASTIC ARTICLES AND SOME SYNTHETIC FABRICS. AVOID TREATING GOGGLES, WATCH CRYSTALS, PAINTED HELMETS, ETC.

Manufacturer's name and address
Date of pack and lot number

5.3.5 Registration of products. The products shall be registered under the Federal Insecticide, Fungicide and Rodenticide Act and be in compliance with its requirements.

6. NOTES

6.1 Intended use. The two types of repellents are intended for direct application on exposed skin surfaces of the body or for application to clothing to protect against mosquitoes, biting flies (to include black, deer, horn, horse, snipe, greenhead, and stable flies), biting midges, gnats, ticks, fleas, and chiggers (mites). Type I will provide a longer lasting repellent action than type II.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number, and date of this specification.
b. Type, solution concentration, and size required (see 1.2.1).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. Size of unit containers for types I and II whether polyethylene bottles or glass bottles are desired (see 5.1.1.1).
e. Type and class of unit load required (see 5.2.1).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
g. When level B packing of intermediate packages is not required for Civil agencies (see 5.2.2.2).

6.3 Standard sample. A standard sample for the N, N-diethyl-m-toluamide in 4.3.1.3 can be obtained from either of the following:

a. Chemical Services Inc.
PO Box 194
West Chester, PA 19380

b. Agricultural Environmental Quality Institute
Beltsville, MD 20705

MILITARY INTERESTS:

Custodians:

Army - GL
Navy - SH
Air Force - 68

Review activities:

Army - MD
DLA - GS

User activities:

Navy - YD, MC, MS

CIVIL AGENCY COORDINATING ACTIVITIES:

USDA - AFS
GSA - FSS
VA - OSS

PREPARING ACTIVITY:

Army - GL

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US Army Natick Research and Development Center
Natick, Massachusetts 01760

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
O-1-503F	Insect Repellent, Clothing and Personal Application

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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